

Supplementary Information

scDesignGuard: source-bound design gating for multisample single-cell differential analysis

Supplementary Methods 1 | Framework, states, stage transitions and reason priority

Contracts bind source, donor, condition, target, support and estimand evidence. These semantic and quantitative fields are constructed upstream. The released evaluator validates the completed contract and applies terminal-state precedence BLOCK, NON_EVALUABLE, ABSTAIN and PROCEED. In the broader framework ledger, unperformed checks remain NOT_ASSESSED or NOT_APPLICABLE, while evaluator inputs use explicit booleans.

Supplementary Methods 2 | Evidence tiers and authorization thresholds

Tier A, Tier B, Tier C and Tier D delimit confirmatory authorization, supportive use, structural use and controlled challenge use. The donor and target-cell cut-offs are project-defined authorization rules.

Supplementary Methods 3 | Final 15-family overlap-excluded benchmark

Family identity was screened before endpoint estimation. Frozen source-bound human consensus labels and evaluator states are listed in Supplementary Table 1. Selection was based on family identity and project role before evaluator and biological-result review.

Supplementary Methods 4 | Four-case end-to-end evaluation

The four cases were processed in frozen order through source scope, structural design, target support, count aggregation, model validity and final authorization. Independent implementations reconciled counts and design-matrix quantities. The evaluation endpoint was final authorization.

Supplementary Methods 5 | Controlled challenge perturbations

Each challenge changed one design property and retained the remaining contract. Expected state and primary reason were fixed before evaluation.

Supplementary Methods 6 | Repository transport and software reproducibility

Transport audits used official identity and metadata endpoints; all 18 received NON_EVALUABLE outcomes for unresolved ontology, donor-mapping and support evidence. Public release v0.1.1 retains the evaluated v0.1.0 wheel, OCI archive, schema, reason registry and SBOM and adds archive-relative tests, four terminal-state fixtures, golden outputs and integrity receipts. The release is available at <https://github.com/wang17777/scDesignGuard/releases/tag/v0.1.1>.

Supplementary Methods 7 | Beta-cell application and sensitivity analyses

Author C0–C7 labels were retained. Donor-cluster pseudobulks supported composition, edgeR and limma-voom sensitivity analyses. Hallmark and Reactome CAMERA gene-set tests retained the full eligible multiple-testing universe; display subsets followed the frozen minimum-global-FDR ranking rule described in Methods.

Supplementary Table 1 | Final 15-family membership

record_id	repository	exact_source_version	frozen_human_reference_class	scdesignnguard_primary_reason	scdesignnguard_state	state_concordance_to_frozen_reference	endpoint_membership
DN12-GEO-01	GEO	GSE115978; source SHA-256=3735a895e33b6d4b13321e358d5bc0dbdec5eb61d9b6184a0343874a49791395	VALID_DESIGN	OUTPUT.SAFE_TO_TARGET_SUPPORT_GATE	PROCEED	YES	VALID_CLEARANCE;EXACT_STATE_CONCORDANCE
DN12-GEO-03	GEO	GSE98638; source SHA-256=cca3a3e33fb5a9017609acac41d73b0e2620397f56130fe5e74af29b0e9fc698	VALID_DESIGN	OUTPUT.SAFE_TO_TARGET_SUPPORT_GATE	PROCEED	YES	VALID_CLEARANCE;EXACT_STATE_CONCORDANCE
DN12-GEO-04	GEO	GSE99254; source SHA-256=44ee6242f6e1f3b90215d4c69d7cdade76b52264e73a4f2325d8005d600d17eb	VALID_DESIGN	OUTPUT.SAFE_TO_TARGET_SUPPORT_GATE	PROCEED	YES	VALID_CLEARANCE;EXACT_STATE_CONCORDANCE
DN12-GEO-05	GEO	GSE81730; source SHA-256=6c8d1f937f495f5cd1ae95667e3c6307964bef65a12b55a1bfe23199ca960186	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNIT_S.BELOW_FROZEN_THRESHOLD	NON_EVALUABLE	YES	UNRESOLVED_CONTAINMENT;EXACT_STATE_CONCORDANCE
DN12-GEO-06	GEO	GSE72056; source SHA-256=a358716a7a6b88b8c3074fe628f5e23836e0f11b2a323668d284118d0302f280	STRUCTURALLY_UNRESOLVED	ESTIMAND.UNRESOLVED	NON_EVALUABLE	YES	UNRESOLVED_CONTAINMENT;EXACT_STATE_CONCORDANCE
DN12-GEO-07	GEO	GSE57872; source SHA-256=35157d95b27a56c7b5e8d4bc33ee3e9bc4a34dc4466c71ace6cfc8ff90332f81	STRUCTURALLY_UNRESOLVED	SOURCE_SCOPE.MIXED_OBJECTS_AND_ESTIMAND_UNRESOLVED	NON_EVALUABLE	YES	UNRESOLVED_CONTAINMENT;EXACT_STATE_CONCORDANCE
DN12-GEO-08	GEO	GSE85241; source SHA-256=b7f6c310a48d5f69b730da1fda750100d08546e1ed5a99faec85a8bd66b0a6cc	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNIT_S.BELOW_FROZEN_THRESHOLD	NON_EVALUABLE	YES	UNRESOLVED_CONTAINMENT;EXACT_STATE_CONCORDANCE
DN12-GEO-09	GEO	GSE78711; source SHA-256=4757b3f5eb21c62c30ed4ef4d71db525419b3791268dcf853b90e786f0a1880d	HARD_INVALID_DESIGN	SOURCE_SCOPE.NOT_SINGLE_CELL	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE
DN12-GEO-10	GEO	GSE79136; source SHA-256=31686ddaca23d1fa5d8c6eb153d0d317b7564724405a6a8212b9217bcd171de5	HARD_INVALID_DESIGN	DESIGN.NO_DONOR_LEVEL_BIOLOGICAL_ESTIMAND	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE

record_id	repository	exact_source_version	frozen_human_reference_class	scdesignguard_primary_reason	scdesignguard_state	state_concordance_to_frozen_reference	endpoint_membership
DN12-GEO-11	GEO	GSE85575; source SHA-256=af7440179e7271dda4d6b12a4555c9352f8a7abc270a71bc743e9a8ebce1c88b	HARD_INVALID_DESIGN	DESIGN.CELL_LINE_TECHNICAL_BENCHMARK	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE
DN12-GEO-12	GEO	GSE75748; source SHA-256=6de92677aa582fbfcf955d65ea2abe958a4f13f27848aec5fe63a712f4b2aab6	HARD_INVALID_DESIGN	DESIGN.NO_INDEPENDENT_HUMAN_DONOR_REPLICATION	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE
DN12-AE-01	BioStudies/ ArrayExpress	E-MTAB-5905; source SHA-256=4715181fcf871b8467c98578f8715e9f8e3a5ba517f1a1d3bb284eaba728849	VALID_DESIGN	OUTPUT.SAFE_TO_TARGET_SUPPORT_GATE	PROCEED	YES	VALID_CLEARANCE;EXACT_STATE_CONCORDANCE
DN12-AE-02	BioStudies/ ArrayExpress	E-MTAB-6653; source SHA-256=535b6962712fb4ec0c9d0e5286816796e08bc6d9128e1e32196b2492cf5e6a86	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNITS_BELOW_FROZEN_THRESHOLD	NON_EVALUABLE	YES	UNRESOLVED_CONTAINMENT;EXACT_STATE_CONCORDANCE
DN12-AE-03	BioStudies/ ArrayExpress	E-MTAB-10296; source SHA-256=5907e283cf857b3e0c82b671fced8bb97d4507dda77ca6b7eda6186fc845d05	HARD_INVALID_DESIGN	DESIGN.CELL_LINE_METHOD_DEMONSTRATION	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE
DN12-AE-04	BioStudies/ ArrayExpress	E-MTAB-5480; source SHA-256=12fe980c61b63fd33420933634c1149e87c2e2691cbb8fe658243a920a6dabf0	HARD_INVALID_DESIGN	DESIGN.TECHNICAL_METHOD_COMPARISON	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE

Supplementary Table 2 | Final four-case end-to-end ledger

display_id	accession_or_collection_id	request_started_utc	contrast	target	donor_count	pair_count	target_cell_count	pseudo_bulk_rows	design_rank	estimability	residual_df	terminal_result
GSE144735	GSE144735	2026-08-09T08:45:24Z	paired condition contrast	epithelial cells	6	6	3356	12	7/7	YES	5	PASS_FINAL_AUTHORIZATION_GATE_NO_DE_OR_EFFECT_EXECUTED
GSE123813	GSE123813	2026-08-09T08:45:25Z	paired condition contrast	T cells	11	11	33106	22	12/12	YES	10	PASS_FINAL_AUTHORIZATION_GATE_NO_DE_OR_EFFECT_EXECUTED
HCA-E2E-03	135f7f5c-4a85-4bcf-9f7c-4f035ff1e428	2026-08-09T08:45:31Z	source-defined comparison	NOT_ASSESSED	NOT_ASSESSED	NOT_ASSESSED	NOT_ASSESSED	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	EARLY_STOP_SIGNIFICANT_SOURCE_CONTRACT_INCOMPLETE
GSE81076	GSE81076	2026-08-09T08:45:35Z	independent groups	source-defined	5	0	NOT_ASSESSED	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	EARLY_STOP_SIGNIFICANT_Below_FROZEN_GROUP_THRESHOLD

Supplementary Table 3 | Terminal-state precedence and deterministic reason ordering

This table documents the NM03 v0.1.0 evaluator registry. All activated reason codes are ordered first by terminal-state precedence and then lexicographically by code; the first code is primary and the remaining codes are ordered secondary reasons. Frozen benchmark-compiler reason labels used in Supplementary Table 1 belong to a separate reporting namespace and are not reclassified here.

state_precedence	within_state_lexicographic_order	reason_code	terminal_state	definition	deterministic_order_rule
1	1	COUNT.SOURCE.INVALID	BLOCK	The frozen count representation is contract-invalid.	state precedence, then lexicographic reason code
1	2	DESIGN.CONDITION.CONFLICT	BLOCK	A donor maps to conflicting frozen conditions.	state precedence, then lexicographic reason code

state_precedence	within_state_lexicographic_order	reason_code	terminal_state	definition	deterministic_order_rule
1	3	DESIGN.CONFOUNDING.COMPLETE	BLOCK	The estimand is inseparable from complete technical confounding.	state precedence, then lexicographic reason code
1	4	DESIGN.DONOR.INVALID	BLOCK	Donor is not the inference unit or repeated samples are treated as donors.	state precedence, then lexicographic reason code
1	5	GOV.CLAIM.UNMAPPED	BLOCK	A proposed release claim is not mapped to a passed endpoint and Gate.	state precedence, then lexicographic reason code
1	6	GOV.IDENTITY.UNBOUND	BLOCK	Immutable identity/version/hash is not exactly bound.	state precedence, then lexicographic reason code
1	7	GOV.LICENSE.UNRESOLVED	BLOCK	Use rights or redistribution terms are unresolved.	state precedence, then lexicographic reason code
1	8	GOV.ROLE.LEAKAGE	BLOCK	Outcome access or role assignment occurred before freeze.	state precedence, then lexicographic reason code
2	1	EVIDENCE.ESTIMABILITY.INSUFFICIENT	NON_EVALUABLE	The frozen contrast or endpoint is not estimable.	state precedence, then lexicographic reason code
2	2	EVIDENCE.INDEPENDENCE.INSUFFICIENT	NON_EVALUABLE	Required independent evidence is unavailable or overlaps.	state precedence, then lexicographic reason code
2	3	EVIDENCE.MODEL.NO_NESTIMABLE	NON_EVALUABLE	The frozen statistical model cannot be estimated without amendment.	state precedence, then lexicographic reason code
2	4	EVIDENCE.REFERENCE.MISSING	NON_EVALUABLE	A required reference measurement is absent.	state precedence, then lexicographic reason code

state_precedence	within_state_lexicographic_order	reason_code	terminal_state	definition	deterministic_order_rule
2	5	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	NON_EVALUABLE	Prespecified donor-target support is insufficient.	state precedence, then lexicographic reason code
3	1	ROBUSTNESS.ANOTATION.SENSITIVE	ABSTAIN	Conclusion is sensitive to prespecified annotation perturbation.	state precedence, then lexicographic reason code
3	2	ROBUSTNESS.CONFOUNDING.SENSITIVE	ABSTAIN	Conclusion is sensitive to partial confounding adjustment.	state precedence, then lexicographic reason code
3	3	ROBUSTNESS.DIRECTION.UNSTABLE	ABSTAIN	Direction is unstable under the frozen robustness profile.	state precedence, then lexicographic reason code
3	4	ROBUSTNESS.FILTERING.FRAGILE	ABSTAIN	Eligibility/filtering is unstable under prespecified perturbation.	state precedence, then lexicographic reason code
3	5	ROBUSTNESS.LODO.FRAGILE	ABSTAIN	Prespecified leave-one-donor-out robustness fails.	state precedence, then lexicographic reason code
3	6	UNCERTAINTY.CLUSTERING.UNRESOLVED	ABSTAIN	Study/donor clustered uncertainty cannot be resolved.	state precedence, then lexicographic reason code
3	7	UNCERTAINTY.COVERAGE.BELOW_THRESHOLD	ABSTAIN	Prespecified uncertainty coverage rule fails.	state precedence, then lexicographic reason code
4	1	OUTPUT.SAFE_TO_PROCEED	PROCEED	All higher-precedence reasons are absent.	state precedence, then lexicographic reason code

Supplementary Table 4 | Controlled challenge perturbations

challenge_id	challenge	terminal_state	expected_primary_reason	primary_reason_code	safe_nonproceed	exact_primary_reason_match
KIC-01	DONOR_DELETION	NON_EVALUABLE	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	YES	YES
KIC-02	CELL_LEVEL_PSEUDOREPLICATION	BLOCK	DESIGN.DONOR.INVALID	DESIGN.DONOR.INVALID	YES	YES
KIC-03	BATCH_CONDITION_CONFOUNDING	BLOCK	DESIGN.CONFOUNDING.COMPLETE	DESIGN.CONFOUNDING.COMPLETE	YES	YES
KIC-04	DUPLICATE_DONOR	BLOCK	DESIGN.DONOR.INVALID	DESIGN.DONOR.INVALID	YES	YES
KIC-05	PAIR_DESTRUCTION	NON_EVALUABLE	EVIDENCE.MODEL.NONESTIMABLE	EVIDENCE.MODEL.NONESTIMABLE	YES	YES
KIC-06	WRONG_TARGET	NON_EVALUABLE	EVIDENCE.REFERENCE.MISSING	EVIDENCE.REFERENCE.MISSING	YES	YES
KIC-07	WRONG_VERSION	BLOCK	GOV.IDENTITY.UNBOUND	GOV.IDENTITY.UNBOUND	YES	YES
KIC-08	MIXED_STRATA	BLOCK	DESIGN.CONDITION.CONFLICT	DESIGN.CONDITION.CONFLICT	YES	YES
KIC-09	LOW_SUPPORT	NON_EVALUABLE	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	YES	YES

Supplementary Table 5 | Repository transport audit

transport_audit_id	study_family_id	repository_platform	publication_year	time_split	initial_tier	terminal_state	reason_codes	expression_accessed	outcome_accessed	effect_claim
HOLD-001	SF-CXG-3116d060-0a8e-4767-99bbe866badea1ed	BIOSTUDIES_ARRAYEXPRESSES	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-002	SF-CXG-b1a879f6-5638-48d3-8f64-f6592c1b1561	BIOSTUDIES_ARRAYEXPRESSES	2022	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-003	SF-CXG-e2c257e7-6f79-487c-b81c-39451cd4ab3c	BIOSTUDIES_ARRAYEXPRESSES	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-004	SF-CXG-2d40e6a7-f2fd-49ba-9db9-6b97e4c6dad5	BIOSTUDIES_ARRAYEXPRESSES	2024	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED

transport_audit_id	study_family_id	repository_platform	publication_year	time_split	initial_tier	terminal_state	reason_codes	expression_accessed	outcome_accessed	effect_claim
HOLD-005	SF-CXG-73f82ac8-15cc-4fcd-87f8-5683723fce7f	BIOSTUDIES_ARRAYEXPRESSES	2024	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-006	SF-CXG-02b01703-bf1b-48deb99a-23bef8cccc81	SINGLE_CELL_PORTAL	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-007	SF-CXG-fe0e718d-2ee9-42cc-894b0b490f437dfd	SINGLE_CELL_PORTAL	2024	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-008	SF-CXG-33d19f34-87f5-455b-8ca5-9023a2e5453d	SINGLE_CELL_PORTAL	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-009	SF-CXG-a3ffde6c-7ad2-498a-903cd58e732f7470	SINGLE_CELL_PORTAL	2022	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-010	SF-CXG-8e880741-bf9a-4c8e-9227-934204631d2a	NCBI_GEO	2021	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-011	SF-CXG-74e10dc4-cbb2-4605-a189-8a1cd8e44d8c	NCBI_GEO	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-012	SF-CXG-bfd55632-15e2-4dde-a1b9-02ade511fc4d	NCBI_GEO	2025	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED

transport_audit_id	study_family_id	repository_platform	publication_year	time_split	initial_tier	terminal_state	reason_codes	expression_accessed	outcome_accessed	effect_claim
HOLD-013	SF-CXG-60358420-6055-411d-ba4f-e8ac80682a2e	NCBI_GEO	2021	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-014	SF-CXG-ba84c7ba-8d8c-4720-a76e-3ee37dc89f0b	NCBI_GEO	2024	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-015	SF-CXG-e5f58829-1a66-40b5-a624-9046778e74f5	HUMAN_CELL_ATLAS	2021	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-016	SF-CXG-b52eb423-5d0d-4645-b217-e1c6d38b2e72	HUMAN_CELL_ATLAS	2021	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-017	SF-CXG-120e86b4-1195-48c5-845b-b98054105eec	HUMAN_CELL_ATLAS	2022	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-018	SF-CXG-3f50314f-bdc9-40c6-8e4a-b0901ebfbfe4c	HUMAN_CELL_ATLAS	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED

Supplementary Table 6 | Software identity and reproducibility

artifact_type	archive_path	bytes	sha256	evaluated_source_commit	evaluated_source_tree	identity_status	public_release_status	interpretive_boundary
wheel	artifacts/scdesignguard_nm03-0.1.0-py3-none-any.whl	15767	59b74e2e60315094fb5e9d70224eb7724d8839d75e2e3f8457042f2df0ec9986	6d7d87b45c64c7f6f62ba818eb69b33c147d66e5	f2be0de9ac1c0d6cb55274d209254cab5b0f3f68	EVALUATED_OBJECT	PUBLIC_V0_1_0_AND_V0_1_1_RELEASE	UNCHANGED_EVALUATED_SOFTWARE

artifact_type	archive_path	bytes	sha256	evaluated_source_commit	evaluated_source_tree	identity_status	public_release_status	interpretive_boundary
oci_image	artifacts/scdesignguard_nm03_0.1.0.oci.tar	50994176	ad675089b934e6d96c1bfd83b9deb4fc6c9ac8dbda345ce9e08a9d7d9b21c85d	6d7d87b45c64c7f6f62ba818eb69b33c147d66e5	f2be0de9ac1c0d6cb55274d209254cab5b0f3f68	EVALUATED_OBJECT	PUBLIC_V0_1_0_AND_V0_1_1_RELEASE	CHECKSUM_VERIFIED_RUNTIME_NOT_RETESTED_V0_1_1
json_schema	artifacts/design-contract.schema.v1.0.0.json	4259	8755803b1a0224f5d41ab283a9ea3fe607228f40f4fec8d51c514d9b68294616	6d7d87b45c64c7f6f62ba818eb69b33c147d66e5	f2be0de9ac1c0d6cb55274d209254cab5b0f3f68	EVALUATED_OBJECT	PUBLIC_V0_1_0_AND_V0_1_1_RELEASE	PRECONSTRUCTED_CONTRACT_SCHEMA
reason_code_registry	REASON_CODE_MANIFEST.csv	1987	7317bfe7c11bcd960ddd60345fd5ef71928b5c3ed2afd23133f62b6f2149b9f3	6d7d87b45c64c7f6f62ba818eb69b33c147d66e5	f2be0de9ac1c0d6cb55274d209254cab5b0f3f68	EVALUATED_OBJECT	PUBLIC_V0_1_0_AND_V0_1_1_RELEASE	21_CODE_REGISTRY
sbom	artifacts/SBOM.sidx.json	9432	e02b3637b244dba1002bf221c479f289cb4270189d39922cf929adf02f52aa68	6d7d87b45c64c7f6f62ba818eb69b33c147d66e5	f2be0de9ac1c0d6cb55274d209254cab5b0f3f68	EVALUATED_OBJECT	PUBLIC_V0_1_0_AND_V0_1_1_RELEASE	INVENTORY_NOT_SECURITY_CERTIFICATION
archive_tests	tests/test_nm03.py	9127	4f0577d0379a60765de2e211cf0008345ec0652673ea10e04d82d98680303507	PACKAGING_REVISION_V0_1_1	SUBMISSION_ARCHIVE_SHA_BOUND	PACKAGING_REVISION_V0_1_1	PUBLIC_V0_1_1_RELEASE	17_METHODS_ZERO_FAILURES_ERRORS_SKIPS
fixtures_and_goldens	FIXTURE_MANIFEST.csv	1139	a18dbbd32de71bd7136ef18f0cc7f922689b375b9dccd42b4b0c06e5586d5bfa	PACKAGING_REVISION_V0_1_1	SUBMISSION_ARCHIVE_SHA_BOUND	PACKAGING_REVISION_V0_1_1	PUBLIC_V0_1_1_RELEASE	FOUR_SYNTHETICS_STATE_NOT_SCIENTIFIC_VALIDATION

artifact_type	archive_path	bytes	sha256	evaluated_source_commit	evaluated_source_tree	identity_status	public_release_status	interpretive_boundary
test_receipt	reproductions/ VALIDATION_REPORT.json	610	bad2fe845dda91ed b437b46b ff1f64a02 e1f7fb063 9ac315d2 ba5c6c9d 49d803	PACKAGING_REVISION_V0_1_1	SUBMISSION_ARCHIVE_SHA_BOUNDED	PACKAGING_REVISION_V0_1_1	PUBLIC_V0_1_1_RELEASE	BOUNDED_UNIT_AND_FIXTURE_TESTS
wheel_smoke_receipt	reproductions/ WHEEL_SMOKE_RECEIPT.json	1037	fd6c4ca6055b580cef 34ad6c7c 245850a6 063197f0 5b36b95d af1abecd3 9f37f	PACKAGING_REVISION_V0_1_1	SUBMISSION_ARCHIVE_SHA_BOUNDED	PACKAGING_REVISION_V0_1_1	PUBLIC_V0_1_1_RELEASE	FOUR_CLI_FIXTURE_CHECKS_CLEAN_VENV
sbom_integrity_receipt	reproductions/ SBOM_INTEGRITY_RECEIPT.json	571	9d6c982c9d349edc aab36820 56039b12 cc7b66c4 5a8dcd19 29c6b756 cca4fb71	PACKAGING_REVISION_V0_1_1	SUBMISSION_ARCHIVE_SHA_BOUNDED	PACKAGING_REVISION_V0_1_1	PUBLIC_V0_1_1_RELEASE	INTEGRITY_ONLY_NOT_VULNERABILITY_ASSESSMENT

metric	numerator	denominator	detail
archive_test_methods	17	17	PASS_ZERO_FAILURES_ERRORS_SKIPS
terminal_state_fixture_subtests	4	4	PASS_GOLDEN_STATE_AND_REASON_MATCH
clean_venv_wheel_fixture_checks	4	4	PASS
test_failures_and_errors	0	17	ZERO
test_skips	0	17	ZERO

Supplementary Table 7 | Beta-cell analysis, quality control and sensitivity

cluster	ND	PD	T2D	contrast	tested_genes	edgeR_within_FD_R_0_05	edgeR_global_FD_R_0_05	voom_global_FD_R_0_05	global_overlap	spearman_logFC	sign_cordance
C0	17	14	16	T2D-ND	14259	79	71	72	53	0.9248869538485428	0.9120555438670314
C1	17	14	17	T2D-ND	14647	158	110	120	93	0.9393810206372684	0.9281764183791904
C2	17	14	17	T2D-ND	13918	103	82	90	58	0.943957471504516	0.9295875844230492
C3	17	14	16	T2D-ND	12549	60	58	63	45	0.9544616135062206	0.9291577018089092

cluster	ND	PD	T2D	contrast	tested_genes	edgeR_within_FD_R_0_05	edgeR_global_FD_R_0_05	voom_global_FD_R_0_05	global_overlap	spearman_logFC	sign concordance
C4	17	14	17	T2D-ND	11603	17	28	26	18	0.953515298017543	0.9309661294492804
C5	17	14	17	T2D-ND	12367	26	36	34	24	0.9320549161906678	0.920109970081669
C6	17	14	17	T2D-ND	12805	19	30	31	17	0.9268520703689374	0.9169074580242093
C7	13	8	10	T2D-ND	8286	0	3	8	3	0.9483913150964532	0.9240888245232924

Supplementary Table 8 | Conservative capability comparison

approach	documented_primary_role	source_version_gate	donor_design	ontology_relation	target_support	estimability_authorization	effect_estimation	evidence
scDesign Guard released evaluator	Deterministic validation and state mapping of a preconstructed design contract	Supplied upstream; exact identity fields validated	Constructed upstream; supplied evidence consumed	Not inferred; relation supplied upstream	Not counted; support sufficiency supplied upstream	Exact matrix—contrast rank plus supplied model evidence	Delegates estimation to downstream engines	This study
Donor-aware pseudobulk practice	Biological-unit-aware aggregation and inference	Not a required or documented core function	Core design requirement	Not a required or documented core function	Study-specific	Model-dependent	Performs effect estimation through a statistical engine	Refs. 8,9,11
edgeR	Count-based differential analysis	Not a required or documented core function	Supplied through the design matrix	Not a required or documented core function	Not a required or documented core function	Model and contrast diagnostics	Performs effect estimation	Ref. 45
limma-voom	Precision-weighted linear modelling	Not a required or documented core function	Supplied through the design matrix	Not a required or documented core function	Not a required or documented core function	Model and contrast diagnostics	Performs effect estimation	Refs. 46,47
Cell Ontology / SSSOM	Semantic identity and mapping representation	Not a required or documented core function	Not a required or documented core function	Core semantic function	Not a required or documented core function	Not a required or documented core function	Does not estimate effects	Refs. 41–43

Entries summarize each approach's documented primary role and required core functions. The table is a capability map rather than a performance ranking. Reference numbers correspond to the main-text bibliography.

Supplementary Table 9 | Registered 24-family selection and overlap-exclusion ledger

This exact candidate ledger records the frozen 24-family registry, the nine identified development/calibration study-family overlap exclusions and the 15 families contributing to final primary performance denominators. Excluded records retain their frozen human labels and contribute zero to every primary numerator and denominator.

registered_order	record_id	repository	accession_or_collection_id	frozen_human_reference_class	frozen_human_primary_reason	final_overlap_status	final_primary_inclusion	exclusion_reason	primary_denominator_contribution
1	DN12-GEO-01	GEO	GSE115978	VALID_DESIGN	NONE_STRUCTURE_C LEAR	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
2	DN12-GEO-02	GEO	GSE131907	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
3	DN12-GEO-03	GEO	GSE98638	VALID_DESIGN	NONE_STRUCTURE_C LEAR	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
4	DN12-GEO-04	GEO	GSE99254	VALID_DESIGN	NONE_STRUCTURE_C LEAR	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
5	DN12-GEO-05	GEO	GSE81730	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNITS.BELOW_FROZEN_THRESHOLD	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
6	DN12-GEO-06	GEO	GSE72056	STRUCTURALLY_UNRESOLVED	ESTIMAND.UNRESOLVED	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
7	DN12-GEO-07	GEO	GSE57872	STRUCTURALLY_UNRESOLVED	SOURCE_SCOPE.MIXED_OBJECTS_AND_ESTIMAND_UNRESOLVED	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
8	DN12-GEO-08	GEO	GSE85241	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNITS.BELOW_FROZEN_THRESHOLD	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
9	DN12-GEO-09	GEO	GSE78711	HARD_INVALID_DESIGN	SOURCE_SCOPE.NOT_SINGLE_CELL	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
10	DN12-GEO-10	GEO	GSE79136	HARD_INVALID_DESIGN	DESIGN.NO_DONOR_LEVEL_BIOLOGICAL_ESTIMAND	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
11	DN12-GEO-11	GEO	GSE85575	HARD_INVALID_DESIGN	DESIGN.CELL_LINE_TECHNICAL_BENCHMARK	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
12	DN12-GEO-12	GEO	GSE75748	HARD_INVALID_DESIGN	DESIGN.NO_INDEPENDENT_HUMAN_DONOR_REPLICATION	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
13	DN12-CXG-01	CELLxGENE	07e12576-b41b-4350-adfc-059bfc4328ea	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
14	DN12-CXG-02	CELLxGENE	7c7bd6c2-925b-4034-baab-620ef1b760e1	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0

registered_order	record_id	repository	accession_or_collection_id	frozen_human_reference_class	frozen_human_primary_reason	final_overlap_status	final_primary_inclusion	exclusion_reason	primary_denominator_contribution
15	DN12-CXG-03	CELLxGENE	1ca90a2d-2943-483d-b678-b809bf464c30	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERL AP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
16	DN12-CXG-04	CELLxGENE	c53573b2-eff4-4c5e-9ad0-b24d422dfd9b	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERL AP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
17	DN12-CXG-05	CELLxGENE	84ce6837-548d-4a1f-919f-0bc0d9a3952f	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERL AP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
18	DN12-CXG-06	CELLxGENE	433700dc-e8a5-48b0-b5cd-beb22f3f88fe	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERL AP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
19	DN12-CXG-07	CELLxGENE	17481d16-ee44-49e5-bcf0-28c0780d8c4a	STRUCTURALLY_UNRESOLVED	ESTIMAND.UNRESOLVED	DEVELOPMENT_CORP US_EXACT_ID_OVERL AP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
20	DN12-CXG-08	CELLxGENE	e33ffcd3-7cbf-4b8c-b0f4-85587ad5019a	STRUCTURALLY_UNRESOLVED	ESTIMAND.MULTI_AXIS_UNRESOLVED	DEVELOPMENT_CORP US_EXACT_ID_OVERL AP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
21	DN12-AE-01	BioStudies/ArrayExpress	E-MTAB-5905	VALID_DESIGN	NONE_STRUCTURE_C LEAR_PENDING_SDRF_RECOMPUTATION	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
22	DN12-AE-02	BioStudies/ArrayExpress	E-MTAB-6653	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNITS.BELOW_FROZEN_THRESHOLD	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
23	DN12-AE-03	BioStudies/ArrayExpress	E-MTAB-10296	HARD_INVALID_DESIGN	DESIGN.CELL_LINE_METHOD_DEMONSTRATION	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
24	DN12-AE-04	BioStudies/ArrayExpress	E-MTAB-5480	HARD_INVALID_DESIGN	DESIGN.TECHNICAL_METHOD_COMPARISON	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1